Lab 4 using your specific repository, layout, and networking assumptions. This version **treats Local subnets as private networks** and **treats the remote server / inter-site network as a 10.10.x.x "internet/public network"** for the purposes of NAT and routing logic. It also assumes the repository hierarchy:

```
None
automation/
├── mockup-infra/
└── week02-stateless-stateful/
    └── phase1-mockup/
```

We will produce:

✅ A revised **high-fidelity lab plan** ✅ **Detailed addressing & topologies** ✅ **Cisco-themed network section** ✅ **IaC / VS Code section** ✅ **Troubleshooting and engineering reflection** ✅ A complete **README.md** ready to be committed to the repo

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🧪 LAB 4 Simulated Internet (10.10.0.0/16) & Private LANs with Stateful vs Stateless Services

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🎯 LAB GOALS

1. **Network Setup:** Create two local labs (A & B) connected via a simulated "public/internet" network (10.10.0.0/16)
2. **NAT + Routing:** Configure static routes and NAT so local traffic reaches remote services
3. **Application Deployment:** Use `mockup-infra` from your repository to deploy stateless and stateful APIs
4. **Testing:** Compare how sessions behave locally vs across the simulated internet

---

👥 TEAM BREAKDOWN (5–6 students)

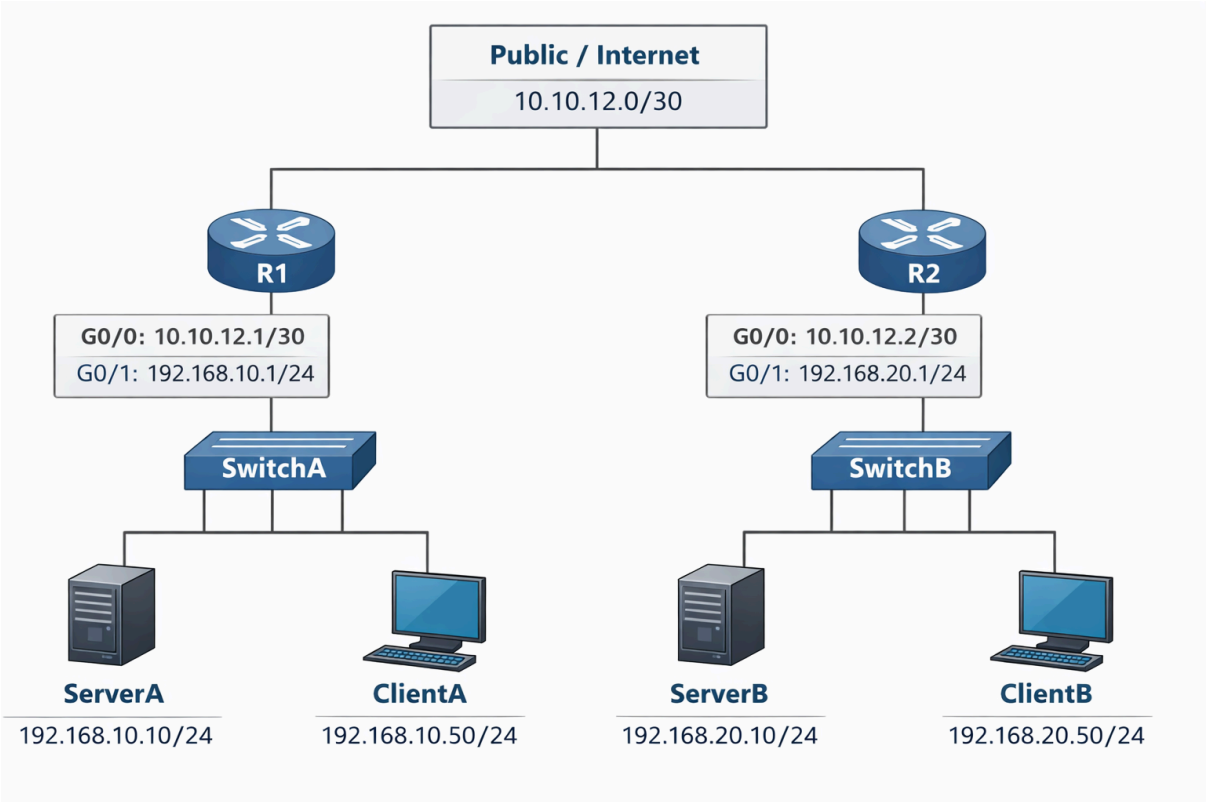
Role	Count	Focus
Network Architects	2	Router configs, Cisco CLI
IaC/DevOps	2	VS Code, mockup-infra

Role	Count	Focus
QA/Test	1–2	Server/Client scripting, session behavior

🗺️ UPDATED NETWORK TOPOLOGY

```python
                10.10.12.0/30 "Public/Internet"
             ┌──────────────────────────┐
             |                          |
         R1-G0/0 (10.10.12.1)      R2-G0/0 (10.10.12.2)
             |                          |
   Private LAN A (192.168.10.0/24)     Private LAN B (192.168.20.0/24)
             |                          |
         SwitchA                      SwitchB
             |                          |
 ServerA (mockup)           ServerB (mockup)
 ClientA                        ClientB
```

We treat 10.10.12.0/30 as a **simulated internet** for the purposes of NAT + public reachability.



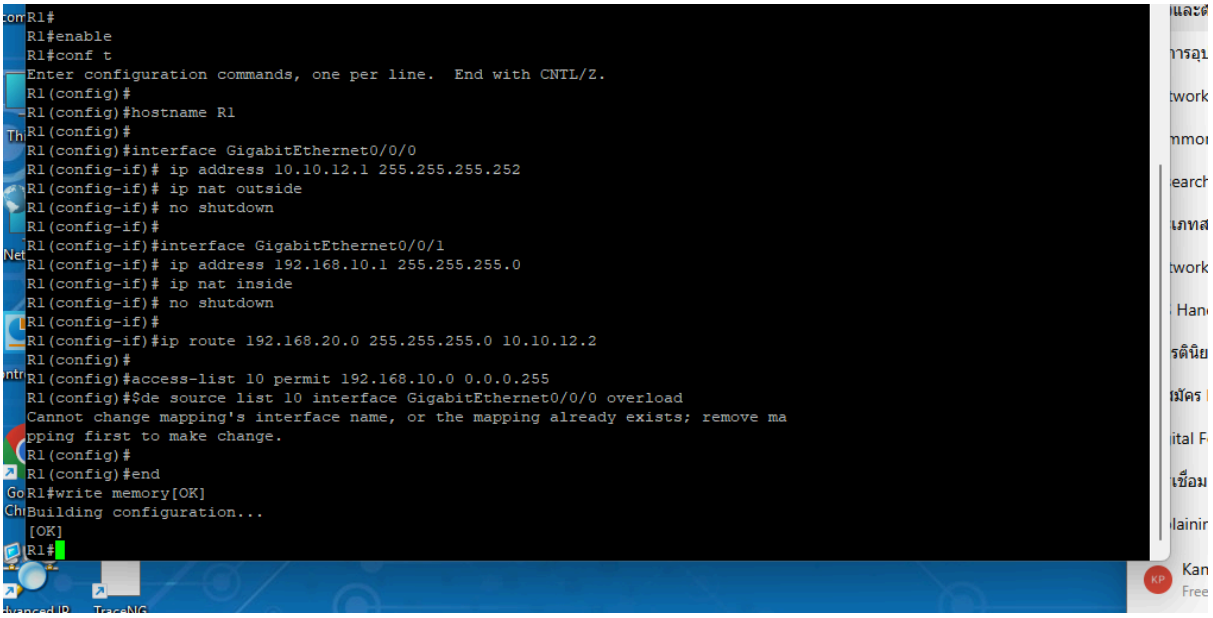
📍 ADDRESS TABLE

Component	Interface	IP	Mask	Role
R1	G0/0	10.10.12.1	/30	Public
R1	G0/1	192.168.10.1	/24	LAN A Gateway
ServerA	eth0	192.168.10.10	/24	Mockup infra A
ClientA	eth0	192.168.10.50	/24	Tester
R2	G0/0	10.10.12.2	/30	Public
R2	G0/1	192.168.20.1	/24	LAN B Gateway
ServerB	eth0	192.168.20.10	/24	Mockup infra B
ClientB	eth0	192.168.20.50	/24	Tester

🔧 PART 1 — CISCO-STYLE NETWORK CONFIGURATION

We will configure static routing + NAT to allow cross-site reachability.

🧱 R1 CONFIGURATION (LAN A)



```
R1#
R1#enable
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#hostname R1
R1(config)#
R1(config)#interface GigabitEthernet0/0/0
R1(config-if)# ip address 10.10.12.1 255.255.255.252
R1(config-if)# ip nat outside
R1(config-if)# no shutdown
R1(config-if)#
R1(config-if)#interface GigabitEthernet0/0/1
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# ip nat inside
R1(config-if)# no shutdown
R1(config-if)#
R1(config-if)#ip route 192.168.20.0 255.255.255.0 10.10.12.2
R1(config)#
R1(config)#access-list 10 permit 192.168.10.0 0.0.0.255
R1(config)#$de source list 10 interface GigabitEthernet0/0/0 overload
Cannot change mapping's interface name, or the mapping already exists; remove ma
pping first to make change.
R1(config)#
R1(config)#end
R1#write memory[OK]
Building configuration...
[OK]
R1#
```

🧱 R2 CONFIGURATION (LAN B)

```
R2>
R2>enable
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#
R2(config)#hostname R2
R2(config)#
R2(config)#interface GigabitEthernet0/0/0
R2(config-if)# ip address 10.10.12.2 255.255.255.252
IP address conflicts with gateway ip address in static routing table
R2(config-if)# ip nat outside
R2(config-if)# no shutdown
R2(config-if)#
R2(config-if)#interface GigabitEthernet0/0/1
R2(config-if)# ip address 192.168.20.1 255.255.255.0
R2(config-if)# ip nat inside
R2(config-if)# no shutdown
R2(config-if)#
R2(config-if)#ip route 192.168.10.0 255.255.255.0 10.10.12.1
%Invalid next hop address (it's this router)
R2(config)#
R2(config)#access-list 10 permit 192.168.20.0 0.0.0.255
R2(config)#$de source list 10 interface GigabitEthernet0/0/0 overload
Cannot change mapping's interface name, or the mapping already exists; remove ma
pping first to make change.
R2(config)#
R2(config)#end
R2#write memory
Building configuration...
[OK]
R2#
```

🧪 ROUTING & NAT VERIFICATION

On R1 & R2:

R1:

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.10.12.0/30 is directly connected, GigabitEthernet0/0/0
L        10.10.12.1/32 is directly connected, GigabitEthernet0/0/0
      192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.10.0/24 is directly connected, GigabitEthernet0/0/1
L        192.168.10.1/32 is directly connected, GigabitEthernet0/0/1
S     192.168.20.0/24 [1/0] via 10.10.12.2
R1#show ip nat translations
Total number of translations: 0

R1#
```

R2 :

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is 10.10.12.2 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 10.10.12.2
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.10.12.0/30 is directly connected, GigabitEthernet0/0/0
L        10.10.12.1/32 is directly connected, GigabitEthernet0/0/0
      192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.20.0/24 is directly connected, GigabitEthernet0/0/1
L        192.168.20.1/32 is directly connected, GigabitEthernet0/0/1
R2#show ip nat translations
Total number of translations: 0

R2#
```

From ClientA:

```
C:\Users\computer>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\computer>
```

💻 PART 2 — IaC + MOCKUP-INFRA DEPLOYMENT (mockup-infra/README.md)

We will now set up mockup infra using the paths you specified:

```
None
automation/mockup-infra
automation/week02-stateless-stateful/phase1-mockup
```

🛠️ CLONE REPO

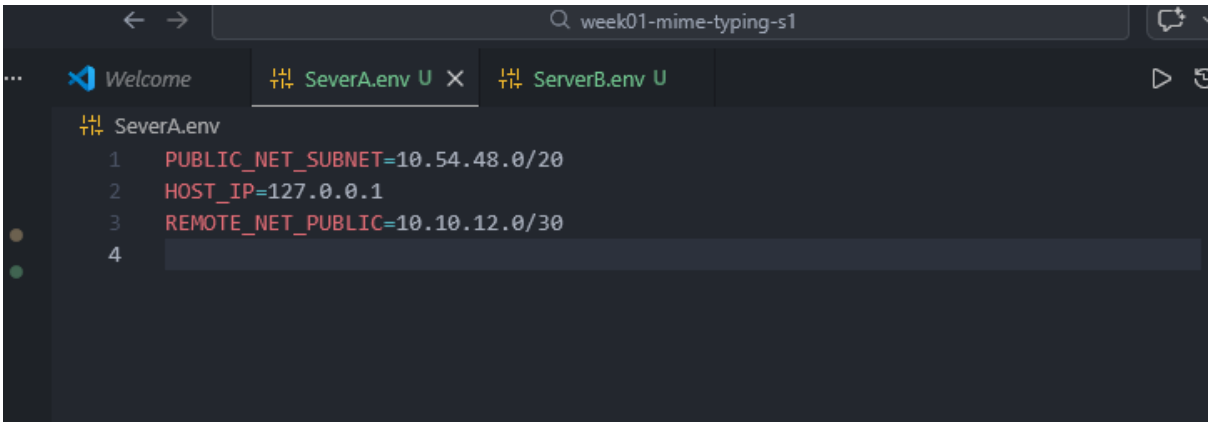
```
└─$ cd lab_04
┌─aekkarin@Penny-2 ~/Desktop/dev/network_lab/lab_04
└─$ git clone https://github.com/boonsup/automation.git
Cloning into 'automation'...
remote: Enumerating objects: 458, done.
remote: Counting objects: 100% (458/458), done.
remote: Compressing objects: 100% (379/379), done.
remote: Total 458 (delta 43), reused 453 (delta 38), pack-reused 0 (from 0)
Receiving objects: 100% (458/458), 730.91 KiB | 1.86 MiB/s, done.
Resolving deltas: 100% (43/43), done.
┌─aekkarin@Penny-2 ~/Desktop/dev/network_lab/lab_04
└─$ ls
automation
┌─aekkarin@Penny-2 ~/Desktop/dev/network_lab/lab_04
└─$ cd automation
┌─aekkarin@Penny-2 ~/Desktop/dev/network_lab/lab_04/automation <main>
└─$
```

🧰 VS CODE — OPEN PROJECT

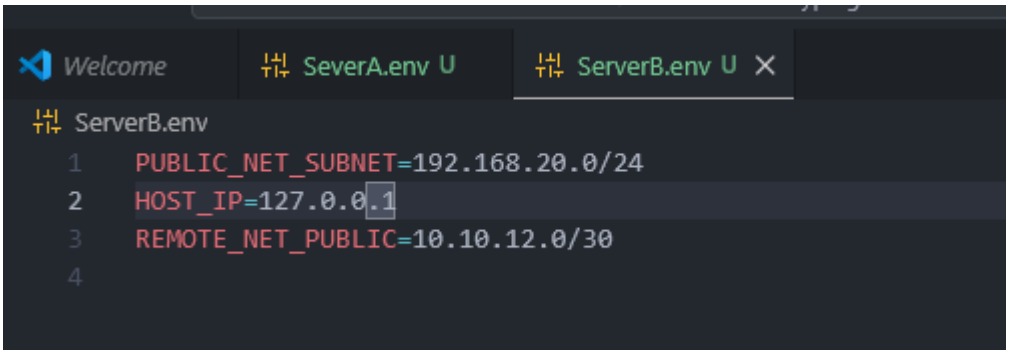
In VS Code, open the `automation` folder.

📄 CREATE SERVER-SPECIFIC ENV FILES

ServerA (`ServerA.env`)



ServerB (`ServerB.env`)



🚀 DEPLOY MOCKUP-INFRA

PowerShell (Windows Admin)

```
HP on D:\..\..\..\mockup-infra main
# python manage.py init

🔧 Initializing Mockup Infrastructure...
==============================================

🔐 [Layer 5/6 - Session] Generating TLS Certificates...
D:\dev\network_lab\automation\mockup-infra\manage.py:118: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future
etimes in UTC: datetime.datetime.now(datetime.UTC).
  .not_valid_before(datetime.utcnow())
D:\dev\network_lab\automation\mockup-infra\manage.py:119: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future
etimes in UTC: datetime.datetime.now(datetime.UTC).
  .not_valid_after(datetime.utcnow() + timedelta(days=365))
✅ Certificates generated successfully:
   - Private Key: D:\dev\network_lab\automation\mockup-infra\certs\server.key
   - Certificate: D:\dev\network_lab\automation\mockup-infra\certs\server.crt
   - Common Name: api.mockup.test
   - Subject Alt Names: localhost, api.mockup.test, 127.0.0.1, 172.18.0.1, 172.19.0.1

🌐 [Layer 3 - Network] Testing Isolation...
🔧 Executing: podman network inspect private_net 2>/dev/null || echo '{}'
'{}'
⚠  Private network not found or not isolated. Creating networks...
🔧 Executing: podman network create public_net 2>/dev/null || true
'true' is not recognized as an internal or external command,
operable program or batch file.
🔧 Executing: podman network create --internal private_net 2>/dev/null || true
'true' is not recognized as an internal or external command,
operable program or batch file.
✅ Networks created successfully

✅ Infrastructure initialized successfully
HP on D:\..\..\..\mockup-infra main
#
```

🟡 VERIFY CONTAINERS

```
HP on D:\..\..\..\..\phase1-mockup main
# podman ps
CONTAINER ID  IMAGE                                       COMMAND               CREATED
    STATUS         PORTS                                       NAMES
6af082bf7b89  localhost/mockup-infra_public_app:latest    python app.py         5 hours ago
    Up 5 hours     80/tcp                                      mockup-public-web
ca6d02c8d42a  localhost/mockup-infra_intranet_api:latest  python api.py         5 hours ago
    Up 5 hours     5000/tcp                                    mockup-intranet-api
ed98c12cee97  docker.io/library/nginx:alpine              nginx -g daemon o...  5 hours ago
    Up 5 hours     0.0.0.0:443->443/tcp, 0.0.0.0:8080->80/tcp  mockup-gateway
a44731ff5118  localhost/lab4-stateless:latest             node start-statel...  24 seconds a
go  Up 25 seconds  0.0.0.0:3000->3000/tcp                      stateless-api
bd2e11ef1515  localhost/lab4-stateful:latest              node start-statef...  19 seconds a
go  Up 20 seconds  0.0.0.0:3001->3001/tcp                      stateful-api
HP on D:\..\..\..\..\phase1-mockup main
#
```

Should show:

Container	Port
stateless-api	3000
stateful-api	3001

🧪 PART 3 — TESTING STATELESS vs STATEFUL (week03-stateless-stateful/phase1-mockup/README.md)

⚫ LOCAL TESTS (Private LAN)

From ClientA:

```
StatusCode        : 200
StatusDescription : OK
Content           : {"status":"success","message":"Stateless server
                    information","data":{"server":"Stateless Mock Server v1.0","timestamp":
                    "2026-03-11T10:34:06.308Z","requestCount":1,"clientId":"anonymous","ran
                    domValue":...
RawContent        : HTTP/1.1 200 OK
                    Content-Security-Policy: default-src 'self';base-uri 'self';font-src
                    'self' https: data:;form-action 'self';frame-ancestors 'self';img-src
                    'self' data:;object-src 'none';script-src 's...
Forms             : {}
Headers           : {[Content-Security-Policy, default-src 'self';base-uri 'self';font-src
                    'self' https: data:;form-action 'self';frame-ancestors 'self';img-src
                    'self' data:;object-src 'none';script-src 'self';script-src-attr
                    'none';style-src 'self' https:
                    'unsafe-inline';upgrade-insecure-requests],
                    [Cross-Origin-Opener-Policy, same-origin],
                    [Cross-Origin-Resource-Policy, same-origin], [Origin-Agent-Cluster,
                    ?1]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 422
```

→ Stateless

From ClientA:

```
HP on D:\..\..\..\..\phase1-mockup main
# curl.exe -X POST http://localhost:3001/session
{"status":"error","message":"Validation failed","errors":[{"field":"body","message":"Missing
 required fields: userId"}],"timestamp":"2026-03-11T10:36:15.641Z"}
HP on D:\..\..\..\..\phase1-mockup main
```

Store the returned `Session-ID`, then:

```shell
curl http://192.168.10.10:3001/data -H "Session-ID: <id>"
```

🔵 REMOTE TESTS (Via Simulated Internet)

From ClientA:

```
# curl.exe http://localhost:3000/info
{"status":"success","message":"Stateless server information","data":{"server":"Stateless Moc
k Server v1.0","timestamp":"2026-03-11T10:39:52.581Z","requestCount":2,"clientId":"anonymous
","randomValue":0.18539100360907024,"message":"I have no memory of previous requests. Each r
equest is independent.","note":"This is a stateless server - I don't remember you from previ
ous requests."},"timestamp":"2026-03-11T10:39:52.581Z"}
HP on D:\..\..\..\..\phase1-mockup main
#
```

That will target **ServerB stateless** — NAT + routing should deliver

---

Session Test:

```
HP on D:\..\..\..\..\phase1-mockup main
# curl.exe -X POST http://localhost:3001/session -H "Content-Type: application/json" -d "{`"
userId`": `"student1`"}"
{"status":"error","message":"Internal server error","errors":[],"timestamp":"2026-03-11T10:4
1:03.388Z"}
HP on D:\..\..\..\..\phase1-mockup main
#
```

→ ServerB returns a session ID

Then:

```
None
curl http://10.10.12.2:3001/data -H "Session-ID: <id>"
```

Check session persists across NAT

⚠️ TROUBLESHOOTING

❗ Cannot Reach Remote

- Gateway misconfiguration
- NAT ACL incorrect
- Wrong default gateway on clients

Check:

```
None
show ip route
ping 10.10.12.2 from R1
```

❗ Sessions Lose State

- Missing header
- Wrong container port
- Firewall blocking port 3001

🧠 ENGINEERING REFLECTION

1. **Stateless services don't depend on prior data**
2. **Stateful services require session headers**
3. NAT shouldn't affect logical session identity

📄 LAB README.md (for repository)

```
None
# Lab 4 — Stateful vs Stateless Across Public & Private Subnets

## Architecture
Two LANs connected via a simulated internet (10.10.12.0/30)
- LAN A: 192.168.10.0/24
- LAN B: 192.168.20.0/24
Routers provide static routing + NAT.

## Steps

### 1. Setup Network
Configure R1 and R2:
- Static routes
- NAT overload
- Verify ping

### 2. Deploy IaC
Configure `ServerA.env` and `ServerB.env`
Deploy via VS Code, PowerShell or Bash using:
```

python manage.py deploy --profile week02-stateless-stateful

```
None

### 3. Validate
- Local stateless vs stateful
- Remote stateless vs stateful
- Test session persistence
- Capture NAT translations

## Troubleshooting
- Check interface status, NAT ACLs
- Verify client gateways

## Extensions
- OSPF instead of static routing
- Load balancer in front of APIs
```